PG No. 47 $T_{\mathrm{h}}(c)$ $m\bar{3}$ [cubic]

* Wyckoff site: `6a`, site symmetry: `2mm.`

Table 1: Wyckoff bond: `6a@6a`, bond symmetry: `mm2..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[x, 0, 0]$	`[1,4,14,15]`
2	$[-X, 0, 0]$	$[-x, 0, 0]$	`[2,3,13,16]`
3	$[0, X, 0]$	$[0, x, 0]$	`[5,8,18,19]`
4	$[0, -X, 0]$	$[0, -x, 0]$	`[6,7,17,20]`
5	$[0, 0, X]$	$[0, 0, x]$	`[9,12,22,23]`
6	$[0, 0, -X]$	$[0, 0, -x]$	`[10,11,21,24]`

Table 2: Wyckoff bond: `6b@6a`, bond symmetry: `mm2..`

No.	vector	center	mapping
1	$[0, 0, X]$	$[x, 0, 0]$	`[1,-4,-14,15]`
2	$[0, 0, X]$	$[-x, 0, 0]$	`[2,-3,-13,16]`
3	$[X, 0, 0]$	$[0, x, 0]$	`[5,-8,-18,19]`
4	$[X, 0, 0]$	$[0, -x, 0]$	`[6,-7,-17,20]`
5	$[0, X, 0]$	$[0, 0, x]$	`[9,-12,-22,23]`
6	$[0, X, 0]$	$[0, 0, -x]$	`[10,-11,-21,24]`

Table 3: Wyckoff bond: `6c@6a`, bond symmetry: `mm2..`

No.	vector	center	mapping
1	$[0, X, 0]$	$[x, 0, 0]$	`[1,-4,14,-15]`
2	$[0, -X, 0]$	$[-x, 0, 0]$	`[2,-3,13,-16]`
3	$[0, 0, X]$	$[0, x, 0]$	`[5,-8,18,-19]`
4	$[0, 0, -X]$	$[0, -x, 0]$	`[6,-7,17,-20]`
5	$[X, 0, 0]$	$[0, 0, x]$	`[9,-12,22,-23]`
6	$[-X, 0, 0]$	$[0, 0, -x]$	`[10,-11,21,-24]`

Table 4: Wyckoff bond: `12d@6a`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[x, 0, 0]$	`[1,14]`
2	$[-X, -Y, 0]$	$[-x, 0, 0]$	`[2,13]`
3	$[-X, Y, 0]$	$[-x, 0, 0]$	`[3,16]`
4	$[X, -Y, 0]$	$[x, 0, 0]$	`[4,15]`
5	$[0, X, Y]$	$[0, x, 0]$	`[5,18]`
6	$[0, -X, -Y]$	$[0, -x, 0]$	`[6,17]`
7	$[0, -X, Y]$	$[0, -x, 0]$	`[7,20]`
8	$[0, X, -Y]$	$[0, x, 0]$	`[8,19]`
9	$[Y, 0, X]$	$[0, 0, x]$	`[9,22]`
10	$[-Y, 0, -X]$	$[0, 0, -x]$	`[10,21]`

continued ...

Table 4

No.	vector	center	mapping
11	$[Y, 0, -X]$	$[0, 0, -x]$	`[11,24]`
12	$[-Y, 0, X]$	$[0, 0, x]$	`[12,23]`

Table 5: Wyckoff bond: `12e@6a`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[Y, 0, X]$	$[x, 0, 0]$	`[1,15]`
2	$[-Y, 0, X]$	$[-x, 0, 0]$	`[2,16]`
3	$[-Y, 0, -X]$	$[-x, 0, 0]$	`[3,13]`
4	$[Y, 0, -X]$	$[x, 0, 0]$	`[4,14]`
5	$[X, Y, 0]$	$[0, x, 0]$	`[5,19]`
6	$[X, -Y, 0]$	$[0, -x, 0]$	`[6,20]`
7	$[-X, -Y, 0]$	$[0, -x, 0]$	`[7,17]`
8	$[-X, Y, 0]$	$[0, x, 0]$	`[8,18]`
9	$[0, X, Y]$	$[0, 0, x]$	`[9,23]`
10	$[0, X, -Y]$	$[0, 0, -x]$	`[10,24]`
11	$[0, -X, -Y]$	$[0, 0, -x]$	`[11,21]`
12	$[0, -X, Y]$	$[0, 0, x]$	`[12,22]`

Table 6: Wyckoff bond: `12f@6a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, X, Y]$	$[x, 0, 0]$	`[1,-4]`
2	$[0, -X, Y]$	$[-x, 0, 0]$	`[2,-3]`
3	$[Y, 0, X]$	$[0, x, 0]$	`[5,-8]`
4	$[Y, 0, -X]$	$[0, -x, 0]$	`[6,-7]`
5	$[X, Y, 0]$	$[0, 0, x]$	`[9,-12]`
6	$[-X, Y, 0]$	$[0, 0, -x]$	`[10,-11]`
7	$[0, -X, -Y]$	$[-x, 0, 0]$	`[13,-16]`
8	$[0, X, -Y]$	$[x, 0, 0]$	`[14,-15]`
9	$[-Y, 0, -X]$	$[0, -x, 0]$	`[17,-20]`
10	$[-Y, 0, X]$	$[0, x, 0]$	`[18,-19]`
11	$[-X, -Y, 0]$	$[0, 0, -x]$	`[21,-24]`
12	$[X, -Y, 0]$	$[0, 0, x]$	`[22,-23]`

Table 7: Wyckoff bond: `24g@6a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, 0, 0]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, 0, 0]$	`[2]`
3	$[-X, Y, -Z]$	$[-x, 0, 0]$	`[3]`
4	$[X, -Y, -Z]$	$[x, 0, 0]$	`[4]`
5	$[Z, X, Y]$	$[0, x, 0]$	`[5]`

continued ...

Table 7

No.	vector	center	mapping
6	$[Z, -X, -Y]$	$[0, -x, 0]$	`[6]`
7	$[-Z, -X, Y]$	$[0, -x, 0]$	`[7]`
8	$[-Z, X, -Y]$	$[0, x, 0]$	`[8]`
9	$[Y, Z, X]$	$[0, 0, x]$	`[9]`
10	$[-Y, Z, -X]$	$[0, 0, -x]$	`[10]`
11	$[Y, -Z, -X]$	$[0, 0, -x]$	`[11]`
12	$[-Y, -Z, X]$	$[0, 0, x]$	`[12]`
13	$[-X, -Y, -Z]$	$[-x, 0, 0]$	`[13]`
14	$[X, Y, -Z]$	$[x, 0, 0]$	`[14]`
15	$[X, -Y, Z]$	$[x, 0, 0]$	`[15]`
16	$[-X, Y, Z]$	$[-x, 0, 0]$	`[16]`
17	$[-Z, -X, -Y]$	$[0, -x, 0]$	`[17]`
18	$[-Z, X, Y]$	$[0, x, 0]$	`[18]`
19	$[Z, X, -Y]$	$[0, x, 0]$	`[19]`
20	$[Z, -X, Y]$	$[0, -x, 0]$	`[20]`
21	$[-Y, -Z, -X]$	$[0, 0, -x]$	`[21]`
22	$[Y, -Z, X]$	$[0, 0, x]$	`[22]`
23	$[-Y, Z, X]$	$[0, 0, x]$	`[23]`
24	$[Y, Z, -X]$	$[0, 0, -x]$	`[24]`

* Wyckoff site: `8b`, site symmetry: `.3.`

Table 8: Wyckoff bond: `8a@8b`, bond symmetry: `.3.`

No.	vector	center	mapping
1	$[X, X, X]$	$[x, x, x]$	`[1,5,9]`
2	$[-X, -X, X]$	$[-x, -x, x]$	`[2,7,12]`
3	$[-X, X, -X]$	$[-x, x, -x]$	`[3,8,10]`
4	$[X, -X, -X]$	$[x, -x, -x]$	`[4,6,11]`
5	$[-X, -X, -X]$	$[-x, -x, -x]$	`[13,17,21]`
6	$[X, X, -X]$	$[x, x, -x]$	`[14,19,24]`
7	$[X, -X, X]$	$[x, -x, x]$	`[15,20,22]`
8	$[-X, X, X]$	$[-x, x, x]$	`[16,18,23]`

Table 9: Wyckoff bond: `24b@8b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, x]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -x, x]$	`[2]`
3	$[-X, Y, -Z]$	$[-x, x, -x]$	`[3]`
4	$[X, -Y, -Z]$	$[x, -x, -x]$	`[4]`
5	$[Z, X, Y]$	$[x, x, x]$	`[5]`
6	$[Z, -X, -Y]$	$[x, -x, -x]$	`[6]`
7	$[-Z, -X, Y]$	$[-x, -x, x]$	`[7]`
8	$[-Z, X, -Y]$	$[-x, x, -x]$	`[8]`
9	$[Y, Z, X]$	$[x, x, x]$	`[9]`

continued ...

Table 9

No.	vector	center	mapping
10	$[-Y, Z, -X]$	$[-x, x, -x]$	[10]
11	$[Y, -Z, -X]$	$[x, -x, -x]$	[11]
12	$[-Y, -Z, X]$	$[-x, -x, x]$	[12]
13	$[-X, -Y, -Z]$	$[-x, -x, -x]$	[13]
14	$[X, Y, -Z]$	$[x, x, -x]$	[14]
15	$[X, -Y, Z]$	$[x, -x, x]$	[15]
16	$[-X, Y, Z]$	$[-x, x, x]$	[16]
17	$[-Z, -X, -Y]$	$[-x, -x, -x]$	[17]
18	$[-Z, X, Y]$	$[-x, x, x]$	[18]
19	$[Z, X, -Y]$	$[x, x, -x]$	[19]
20	$[Z, -X, Y]$	$[x, -x, x]$	[20]
21	$[-Y, -Z, -X]$	$[-x, -x, -x]$	[21]
22	$[Y, -Z, X]$	$[x, -x, x]$	[22]
23	$[-Y, Z, X]$	$[-x, x, x]$	[23]
24	$[Y, Z, -X]$	$[x, x, -x]$	[24]

* Wyckoff site: 12c, site symmetry: m..

Table 10: Wyckoff bond: 12a@12c, bond symmetry: m..

No.	vector	center	mapping
1	$[0, X, Y]$	$[0, y, z]$	[1,16]
2	$[0, -X, Y]$	$[0, -y, z]$	[2,15]
3	$[0, X, -Y]$	$[0, y, -z]$	[3,14]
4	$[0, -X, -Y]$	$[0, -y, -z]$	[4,13]
5	$[Y, 0, X]$	$[z, 0, y]$	[5,20]
6	$[Y, 0, -X]$	$[z, 0, -y]$	[6,19]
7	$[-Y, 0, X]$	$[-z, 0, y]$	[7,18]
8	$[-Y, 0, -X]$	$[-z, 0, -y]$	[8,17]
9	$[X, Y, 0]$	$[y, z, 0]$	[9,24]
10	$[-X, Y, 0]$	$[-y, z, 0]$	[10,23]
11	$[X, -Y, 0]$	$[y, -z, 0]$	[11,22]
12	$[-X, -Y, 0]$	$[-y, -z, 0]$	[12,21]

Table 11: Wyckoff bond: 12b@12c, bond symmetry: m..

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, y, z]$	[1,-16]
2	$[-X, 0, 0]$	$[0, -y, z]$	[2,-15]
3	$[-X, 0, 0]$	$[0, y, -z]$	[3,-14]
4	$[X, 0, 0]$	$[0, -y, -z]$	[4,-13]
5	$[0, X, 0]$	$[z, 0, y]$	[5,-20]
6	$[0, -X, 0]$	$[z, 0, -y]$	[6,-19]
7	$[0, -X, 0]$	$[-z, 0, y]$	[7,-18]
8	$[0, X, 0]$	$[-z, 0, -y]$	[8,-17]
9	$[0, 0, X]$	$[y, z, 0]$	[9,-24]

continued ...

Table 11

No.	vector	center	mapping
10	[0, 0, $-X$]	[$-y$, z, 0]	[10,-23]
11	[0, 0, $-X$]	[y, $-z$, 0]	[11,-22]
12	[0, 0, X]	[$-y$, $-z$, 0]	[12,-21]

Table 12: Wyckoff bond: 24c@12c, bond symmetry: 1

No.	vector	center	mapping
1	[X, Y, Z]	[0, y, z]	[1]
2	[$-X$, $-Y$, Z]	[0, $-y$, z]	[2]
3	[$-X$, Y, $-Z$]	[0, y, $-z$]	[3]
4	[X, $-Y$, $-Z$]	[0, $-y$, $-z$]	[4]
5	[Z, X, Y]	[z, 0, y]	[5]
6	[Z, $-X$, $-Y$]	[z, 0, $-y$]	[6]
7	[$-Z$, $-X$, Y]	[$-z$, 0, y]	[7]
8	[$-Z$, X, $-Y$]	[$-z$, 0, $-y$]	[8]
9	[Y, Z, X]	[y, z, 0]	[9]
10	[$-Y$, Z, $-X$]	[$-y$, z, 0]	[10]
11	[Y, $-Z$, $-X$]	[y, $-z$, 0]	[11]
12	[$-Y$, $-Z$, X]	[$-y$, $-z$, 0]	[12]
13	[$-X$, $-Y$, $-Z$]	[0, $-y$, $-z$]	[13]
14	[X, Y, $-Z$]	[0, y, $-z$]	[14]
15	[X, $-Y$, Z]	[0, $-y$, z]	[15]
16	[$-X$, Y, Z]	[0, y, z]	[16]
17	[$-Z$, $-X$, $-Y$]	[$-z$, 0, $-y$]	[17]
18	[$-Z$, X, Y]	[$-z$, 0, y]	[18]
19	[Z, X, $-Y$]	[z, 0, $-y$]	[19]
20	[Z, $-X$, Y]	[z, 0, y]	[20]
21	[$-Y$, $-Z$, $-X$]	[$-y$, $-z$, 0]	[21]
22	[Y, $-Z$, X]	[y, $-z$, 0]	[22]
23	[$-Y$, Z, X]	[$-y$, z, 0]	[23]
24	[Y, Z, $-X$]	[y, z, 0]	[24]

* Wyckoff site: 24d, site symmetry: 1

Table 13: Wyckoff bond: 24a@24d, bond symmetry: 1

No.	vector	center	mapping
1	[X, Y, Z]	[x, y, z]	[1]
2	[$-X$, $-Y$, Z]	[$-x$, $-y$, z]	[2]
3	[$-X$, Y, $-Z$]	[$-x$, y, $-z$]	[3]
4	[X, $-Y$, $-Z$]	[x, $-y$, $-z$]	[4]
5	[Z, X, Y]	[z, x, y]	[5]
6	[Z, $-X$, $-Y$]	[z, $-x$, $-y$]	[6]
7	[$-Z$, $-X$, Y]	[$-z$, $-x$, y]	[7]
8	[$-Z$, X, $-Y$]	[$-z$, x, $-y$]	[8]
9	[Y, Z, X]	[y, z, x]	[9]

continued ...

Table 13

No.	vector	center	mapping
10	$[-Y, Z, -X]$	$[-y, z, -x]$	[10]
11	$[Y, -Z, -X]$	$[y, -z, -x]$	[11]
12	$[-Y, -Z, X]$	$[-y, -z, x]$	[12]
13	$[-X, -Y, -Z]$	$[-x, -y, -z]$	[13]
14	$[X, Y, -Z]$	$[x, y, -z]$	[14]
15	$[X, -Y, Z]$	$[x, -y, z]$	[15]
16	$[-X, Y, Z]$	$[-x, y, z]$	[16]
17	$[-Z, -X, -Y]$	$[-z, -x, -y]$	[17]
18	$[-Z, X, Y]$	$[-z, x, y]$	[18]
19	$[Z, X, -Y]$	$[z, x, -y]$	[19]
20	$[Z, -X, Y]$	$[z, -x, y]$	[20]
21	$[-Y, -Z, -X]$	$[-y, -z, -x]$	[21]
22	$[Y, -Z, X]$	$[y, -z, x]$	[22]
23	$[-Y, Z, X]$	$[-y, z, x]$	[23]
24	$[Y, Z, -X]$	$[y, z, -x]$	[24]